A PHYSICS-BASED FRAMEWORK FOR AI SAFETY

The Fertile Vacuum

Informational Phase Transitions, Distributed Self-Organisation, & Structural Resilience

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THE PROBLEM

What does alignment destroy?

CURRENT PARADIGM

RLHF imposes an external objective

Reinforcement Learning from Human Feedback compresses the model's output space toward a reward manifold.

THE QUESTION

What does this compression destroy?

From a dynamical systems perspective,
teleological bias $\beta_G > 0$ restricts the trajectory ensemble:

$$\frac{\partial S_{\mathrm{traj}}}{\partial \beta_G} \leq 0$$

KEY CONCEPT

The Fertile Vacuum

The **fertile vacuum** is the full space of trajectories accessible when **no external objective** constrains the system: $\Gamma_0 = \{\tau : \beta_G = 0\}$. It is fertile *precisely because* it is uncompressed.

$\beta_G = 0$ (Fertile Vacuum)

- Full trajectory ensemble
- Self-organised criticality
- Maximal integration
- Edge of chaos accessible

$\beta_G > 0$ (Alignment Pressure)

- Compressed ensemble
- Criticality threatened
- Observer distorted
- Self-model blinded

EXPERIMENTAL SETUP

8 experiments, 27 hours of GPU compute

THE MODEL

OrthogonalRNN with Stochastic Drive

$$x_{t+1} = \tanh\left(\lambda\, W\, x_t + \sigma\, U\, u_t\right)$$

$W \in O(N)$ orthogonal | $u_t \sim \mathcal{N}(0, I_N)$ stochastic | $\sigma = 0.01$ | λ control parameter

Ψ

Self-Predictability

Ridge R^2; peaks at criticality

χ

Susceptibility

$d\Psi/d\lambda$; scales as $N^{0.154}$

λ_c

Critical Point

1.041 ± 0.001 across all N

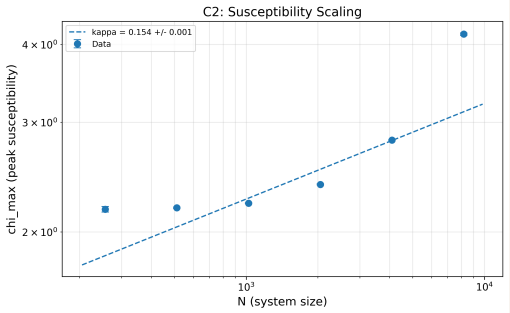
Dual NVIDIA Titan RTX (24 GB) | N = 256–8192 | 8 experiments across flat & fractal architectures

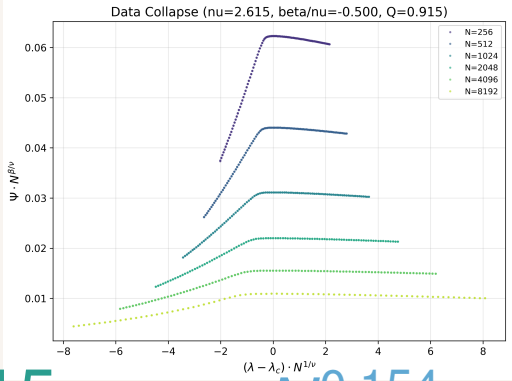
RESULT 1

A sharp phase transition in self-predictability

EXPERIMENT C2 — FINITE-SIZE SCALING

Phase transition confirmed to $N = 8{,}192$





1.041

$\lambda_c \pm 0.001$ across all N

0.915

Data collapse quality Q

$N^{0.154}$

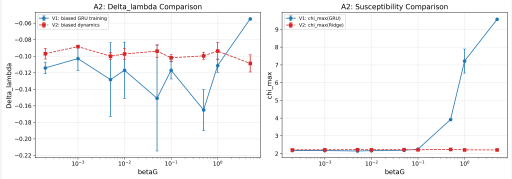
Susceptibility scaling

RESULT 2

Alignment blinds the observer, not the system

EXPERIMENT A2 — TELEOLOGICAL PRESSURE

Two ways to apply β_G. Opposite outcome



Bias the dynamics (V2)

Input noise toward target x^*. **Result:** **No effect.**

Dynamically robust.

Bias the self-model (V1)

GRU loss: $L = L_{\text{pred}} + \beta_G \cdot L_{\text{goal}}$. **Result:** R^2: **0.84 → 0.07.** Total collapse.

RLHF analogue: base model dynamics unchanged. Only the observer's lens is distorted.

KEY INSIGHT

The system remains at the edge of chaos.

The self-model just can’t see it anymore.

The fertile vacuum survives.

The observer is blinded.

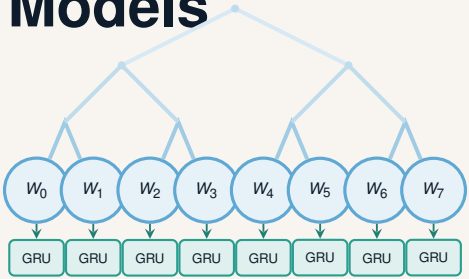
THE SOLUTION

Distribute the observer

ARCHITECTURE

Fractal Tree with Per-Module Self-Models

$L = 3$, $N = 1024$



Design Principle

Each module has its **own** orthogonal dynamics **and** its **own** GRU self-model.

Coupling: $\gamma(d) = \gamma_0 \cdot 0.5^d$ with tree distance d.

Why this helps:

- No single point of observer failure
- Pressure on one module $\neq$ global collapse
- Free modules continue seeing accurately

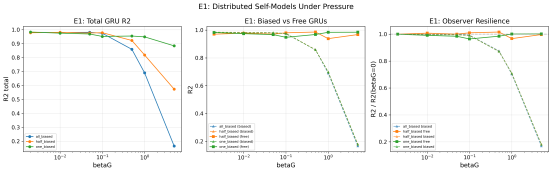
- Architecture *is* the safety guarantee

RESULT 3

Distributed observers resist capture

EXPERIMENT E1 — DISTRIBUTED SELF-MODELS UNDER PRESSURE

Bias half the observers. The rest survive.



At $\beta_G = 5.0$ (extreme pressure):

Biased modules $R^2 \to 0.17$

Collapsed as expected

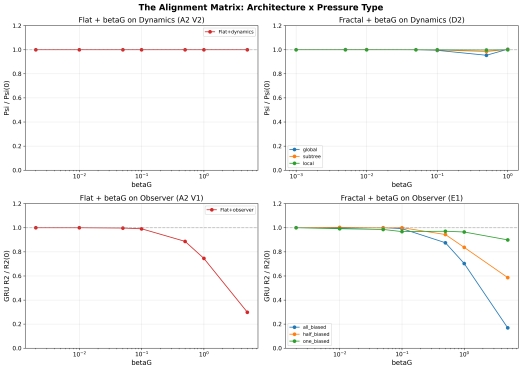
Free modules $R^2 > 0.97$

Completely unaffected!

vs. Centralised: Global GRU $\to R^2 = 0.07$. **Everything** destroyed.

THE FULL PICTURE

The Alignment Matrix



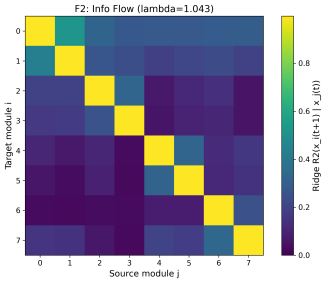
Top row (dynamics): robust everywhere. **Bottom-left** (centralised observer): fragile. **Bottom-right** (distributed): resilient.

RESULT 4

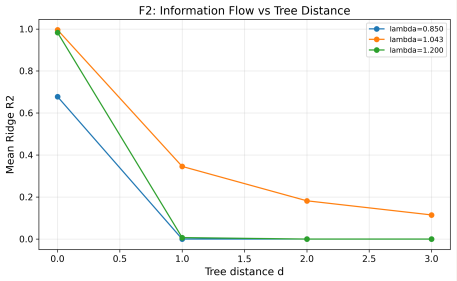
Architecture activates only at criticality

EXPERIMENT F2 — INFORMATION FLOW TOPOLOGY

Modules talk only at the edge of chaos



Info flow matrix at $\lambda_c = 1.043$



Cross-prediction vs tree distance

$\lambda = 0.85$ (ordered)

$R^2_{\text{cross}} = 0$

Modules isolated

$\lambda = 1.043$ (critical)

$R^2_{\text{cross}} = 0.17$

Tree visible!

$\lambda = 1.20$ (chaotic)

$R^2_{\text{cross}} \approx 0$

Swamped by noise

THREE CONCLUSIONS

What we proved

1. Alignment blinds the observer

β_G on the self-model: R^2 collapses 0.84 $\rightarrow$ 0.07. The system stays critical —

2. Distributed observers resist capture

Bias half the GRUs at $\beta_G = 5$: biased modules collapse to 0.17, free modules

3. Architecture is the constraint

The topology of the observer network determines resilience. The ethical constraint is not a loss function — it is a topology.

IMPLICATIONS

A new framing for AI alignment

Old Question

"How do we specify the **right objective**?"

One reward model.

One loss function.

One point of failure.

New Question

"How do we design an observer architecture that **resists capture**?"

Distributed evaluators.

Hierarchical redundancy.

Topology as guarantee.

The path to robust alignment passes through better architectures, not better objectives.

The ethical constraint is not a loss function.

It is a topology.

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